A 2.5 GHz Microstrip Patch Antenna: Transmission-Line Synthesis and Full-Wave Optimisation

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Report type	Technical report
Status	Design and simulation complete. Validation is numerical; no fabricated prototype is reported
Tools	CST Studio Suite (frequency-domain solver), Python
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Abstract. A rectangular inset-fed microstrip patch antenna was designed for 2.5 GHz on FR-4 (ε_r = 4.6, h = 1.6 mm). Closed-form transmission-line expressions gave the initial geometry, which was then simulated and optimised in CST with the frequency-domain solver. A mesh-convergence study fixed the discretisation at 30 lines per wavelength. Parameter sweeps showed that patch length primarily sets resonant frequency (−64 MHz/mm) and inset depth primarily sets the impedance match. The optimised design has a simulated S_{11} of −24.3 dB at 2.5 GHz, 5.8 dBi realised gain, 7.2 dBi directivity and 72% radiation efficiency. The optimised patch length (27.8 mm) differs from the analytical value (27.6 mm) by 0.7%, supporting the transmission-line model as an initial design method under the simulated FR-4 assumptions. No measurement of a fabricated antenna was performed.

Index terms: microstrip patch antenna, transmission-line model, inset feed, CST, mesh convergence, FR-4.

1. Objectives and Design Targets

The objective was to design a patch antenna for 2.5 GHz using an analytical model to set the starting geometry and a full-wave solver to refine it, and to quantify the agreement between the two. FR-4 was chosen as a low-cost substrate despite its loss tangent.

Table 1. Design targets.

Target	Value
Operating frequency	2.5 GHz
Input impedance	50 Ω
Return loss at resonance	$S_{11} < -20$ dB
Substrate	FR-4, ε_r = 4.6, tanδ = 0.015, h = 1.6 mm
Board	60 x 60 mm, 35 µm copper

2. Analytical Synthesis

The transmission-line model treats the patch as a low-impedance line between two radiating slots [1], [3]. Patch width, effective permittivity and resonant length are

$$W_p = c / (2f_r) \cdot \sqrt{(2 / (\varepsilon_r + 1))} \quad (1)$$

$$\varepsilon_{eff} = (\varepsilon_r + 1)/2 + (\varepsilon_r - 1)/2 \cdot (1 + 12h / W_p)^{-1/2} \quad (2)$$

$$L_p = c / (2f_r\sqrt{\varepsilon_{eff}}) - 2\Delta L \quad (3)$$

where ΔL is the fringing-field extension. The radiating-edge resistance (taken as 200 Ω) is matched to 50 Ω with an inset feed, using $R_{in}(y_0) = R_{edge} \cos^2(\pi y_0/L_p)$. The 50 Ω feed width follows from the Hammerstad equations [4].

Table 2. Analytically synthesised geometry (initial CST model).

Parameter	Symbol	Value	Evidence
Feed width	W_f	3.0 mm	Calculated
Patch width	W_p	35.9 mm	Calculated
Effective permittivity	ε_{eff}	4.25	Calculated
Patch length	L_p	27.6 mm	Calculated
Inset depth	y_0	9.2 mm (9.0 mm used in the initial model)	Calculated

3. Simulation Method

The model (PEC ground plane, FR-4 substrate, 3.0 mm feed line and inset-fed patch) was solved with the CST frequency-domain tetrahedral solver from 2.0 to 3.0 GHz [5]. Open (add-space) boundaries on ±X, ±Y and +Z model radiation into free space; an electric boundary at Z-min represents the ground plane. The waveguide port was sized at five times the feed width and six times the substrate height to capture the fringing fields.

3.1 Mesh convergence

Mesh density was increased until the per-step change in resonant frequency and return loss fell below 0.5%, which occurred between 25 and 30 lines per wavelength. A density of 30 lines per wavelength was used for all subsequent sweeps (Table 3, Figure 1). The study was performed on the initial, pre-optimisation geometry.

Table 3. Mesh-convergence data (simulated).

Lines/λ	Cells	f_r (GHz)	S_{11} (dB)
10	12.5k	2.62	−12.4
20	89.6k	2.54	−16.2
30	285k	2.51	−17.2
40	648k	2.50	−17.4

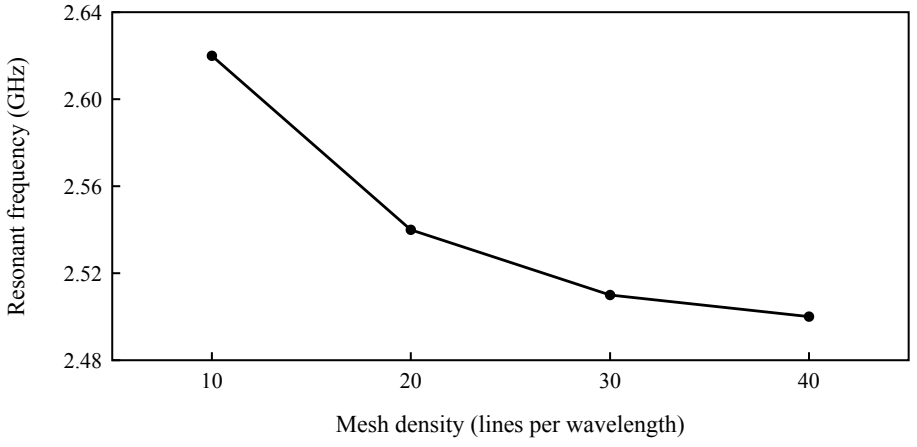


Figure 1. Simulated resonant frequency against mesh density for the initial geometry. Between 30 and 40 lines per wavelength f_r changes by 0.01 GHz and S_{11} by 0.2 dB.

4. Optimisation

A patch-length sweep gave an approximately linear frequency sensitivity of −64 MHz/mm; interpolation placed resonance at 2.50 GHz for L_p = 27.8 mm. An inset-depth sweep changed the match with little effect on frequency; the best match, −24.3 dB, occurred at y_0 = 9.5 mm. Because the two parameters act largely independently, they could be tuned sequentially.

5. Results

Table 4. Final performance.

Quantity	Value	Evidence
S_{11} at 2.5 GHz	−24.3 dB	Simulated
Peak realised gain	5.8 dBi	Simulated
Peak directivity	7.2 dBi	Simulated
Radiation efficiency	72%	Simulated
E-plane half-power beamwidth	78°	Simulated
H-plane half-power beamwidth	96°	Simulated
Front-to-back ratio	18.5 dB	Simulated
Optimised patch length	27.8 mm	Simulated (optimisation)
Analytical patch length	27.6 mm	Calculated
Length agreement	0.7%	Derived
Optimised inset depth	9.5 mm (analytical 9.2 mm)	Simulated

6. Discussion

Model agreement. The 0.7% difference in patch length indicates that the transmission-line model provides an effective starting point; optimisation was a refinement rather than a search. The inset depth differed more (9.5 mm against 9.2 mm), consistent with the approximate edge-resistance value used in the $\cos^2$ relation.

Losses. The 1.4 dB difference between directivity and realised gain reflects radiation efficiency (72%) and mismatch. On FR-4, dielectric loss ($\tan\delta = 0.015$) is expected to dominate, with conductor and surface-wave contributions [1]. The substrate is electrically thin ($h \approx 0.013\lambda_0$), which limits surface-wave excitation.

Bandwidth. Fractional bandwidth follows from the loaded quality factor, $FBW = (VSWR - 1) / (Q\sqrt{VSWR})$. For a thin FR-4 patch the VSWR < 2 bandwidth is expected to be a few per cent of the centre frequency; this was estimated, not extracted from the simulation.

7. Limitations

- Validation is numerical only. No antenna was fabricated or measured, so S_{11} and pattern results are unconfirmed experimentally.
- FR-4 permittivity, loss tangent and thickness vary between batches; their effect on f_r, S_{11} and efficiency was not swept.
- Bandwidth and cross-polarisation were not quantified.

8. Conclusion and Future Work

The design met the −20 dB return-loss target in simulation with a margin of 4.3 dB, and the analytical and optimised lengths agreed to 0.7%. Future work: fabricate and measure S_{11} with a vector network analyser; run a sensitivity analysis over ε_r, h and fabrication tolerance; evaluate a low-loss laminate such as Rogers RO4003; and investigate bandwidth enhancement and arraying.

Appendix A. Synthesis Script

```
import numpy as np
c, fr, er, h = 3e8, 2.5e9, 4.6, 1.6e-3
Wp   = c/(2*fr) * np.sqrt(2/(er+1))
eeff = (er+1)/2 + (er-1)/2 * (1 + 12*h/Wp)**-0.5
dL   = 0.412*h * ((eeff+0.3)*(Wp/h+0.264)) / ((eeff-0.258)*(Wp/h+0.8))
Lp   = c/(2*fr*np.sqrt(eeff)) - 2*dL
y0   = (Lp/np.pi) * np.arccos(np.sqrt(50.0/200.0))   # R_edge = 200 ohm
# Wp = 35.9 mm, eeff = 4.25, Lp = 27.6 mm, y0 = 9.2 mm
```

Listing A1. Closed-form synthesis of the initial geometry.

References

[1] C. A. Balanis, *Antenna Theory: Analysis and Design*, 4th ed. Wiley, 2016.

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[5] Dassault Systèmes, "CST Studio Suite: frequency-domain solver," documentation.